

CO4 AT3 PYTHON INDUSTRY PROBLEM SOLVING TASK

CODE:

```
main.py
1 from datetime import datetime
2
3 class Question:
4     def __init__(self, question, options, answer):
5         self.question = question
6         self.options = options
7         self.answer = answer
8
9 class Quiz:
10     def __init__(self, title):
11         self.title = title
12         self.questions = []
13
14     def add_question(self, question):
15         self.questions.append(question)
16
17 class StudentResponse:
18     def __init__(self, name):
19         self.name = name
20         self.answers = []
21         self.score = 0
22         self.correct = 0
23         self.wrong = 0
24         self.missing = 0
25
26     def add_answer(self, answer):
27         self.answers.append(answer)
28
29     def evaluate(self, questions):
30         for i in range(len(questions)):
31             answer = self.answers[i]
32
33             if answer == "":
```

```
main.py
33 ~         if answer == "":
34 ~             self.missing += 1
35 ~         elif answer == questions[i].answer:
36 ~             self.correct += 1
37 ~             self.score += 1
38 ~         else:
39 ~             self.wrong += 1
40
41 ~     def report(self, total):
42 ~         percentage = (self.score / total) * 100
43
44 ~         if percentage >= 90:
45 ~             grade = "A+"
46 ~         elif percentage >= 80:
47 ~             grade = "A"
48 ~         elif percentage >= 70:
49 ~             grade = "B"
50 ~         elif percentage >= 60:
51 ~             grade = "C"
52 ~         elif percentage >= 50:
53 ~             grade = "D"
54 ~         else:
55 ~             grade = "F"
56
57 ~         print("\n===== RESULT REPORT =====")
58 ~         print("Student Name:", self.name)
59 ~         print("Date:", datetime.now().strftime("%d-%m-%Y %H:%M:%S"))
60 ~         print("Total Questions:", total)
61 ~         print("Correct Answers:", self.correct)
62 ~         print("Wrong Answers:", self.wrong)
63 ~         print("Missing Answers:", self.missing)
64 ~         print("Score:", self.score, "/", total)
65 ~         print("Percentage:", format(percentage, ".2f") + "%")
```

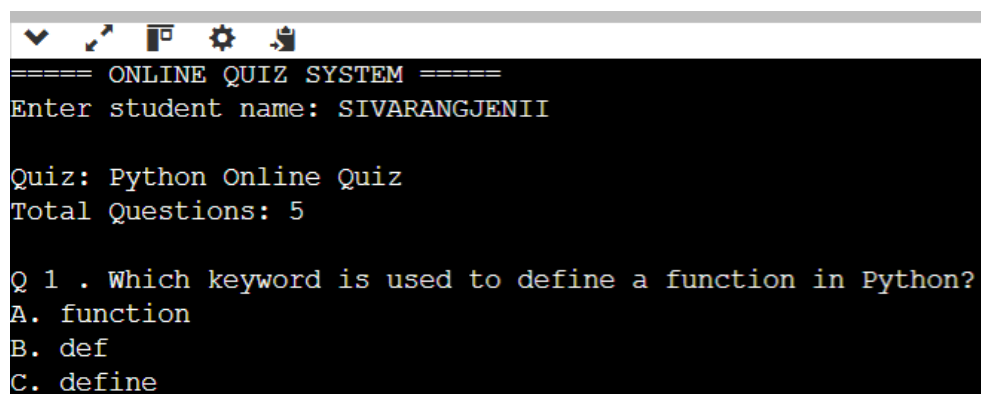
```
65 ~         print("Percentage:", format(percentage, ".2f") + "%")
66 ~         print("Grade:", grade)
67 ~         print("=====")
68
69 ~     quiz = Quiz("Python Online Quiz")
70
71 ~     quiz.add_question(Question(
72 ~         "Which keyword is used to define a function in Python?",
73 ~         ["A. function", "B. def", "C. define", "D. fun"], "B"))
74
75 ~     quiz.add_question(Question(
76 ~         "Which data type stores multiple values in Python?",
77 ~         ["A. list", "B. int", "C. float", "D. bool"], "A"))
78
79 ~     quiz.add_question(Question(
80 ~         "Which symbol is used for comments in Python?",
81 ~         ["A. //", "B. /*", "C. #", "D. --"], "C"))
82
83 ~     quiz.add_question(Question(
84 ~         "Which function displays output in Python?",
85 ~         ["A. input()", "B. print()", "C. display()", "D. output()"], "B"))
86
87 ~     quiz.add_question(Question(
88 ~         "Which keyword is used to loop through a sequence?",
89 ~         ["A. repeat", "B. loop", "C. for", "D. next"], "C"))
90
91 ~     print("===== ONLINE QUIZ SYSTEM =====")
92
93 ~     name = input("Enter student name: ")
94 ~     student = StudentResponse(name)
95
96 ~     print("\nQuiz:", quiz.title)
97 ~     print("Total Questions:", len(quiz.questions))
```

```

96 print("\nQuiz: ", quiz.title)
97 print("Total Questions:", len(quiz.questions))
98
99 for i, q in enumerate(quiz.questions):
100     print("\nQ", i + 1, ".", q.question)
101
102     for option in q.options:
103         print(option)
104
105     answer = input("Enter your answer (A/B/C/D): ").strip().upper()
106
107     while answer not in ["A", "B", "C", "D", ""]:
108         print("Invalid answer! Please enter A, B, C or D.")
109         answer = input("Enter your answer: ").strip().upper()
110
111     student.add_answer(answer)
112
113 student.evaluate(quiz.questions)
114 student.report(len(quiz.questions))
115
116 print("\nQuiz completed successfully!")

```

INPUT & OUTPUT:



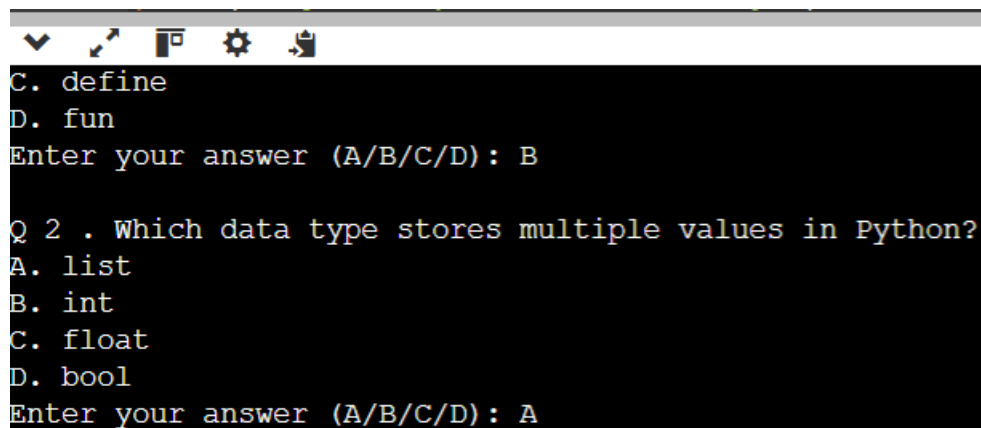
```

==== ONLINE QUIZ SYSTEM ====
Enter student name: SIVARANGJENII

Quiz: Python Online Quiz
Total Questions: 5

Q 1 . Which keyword is used to define a function in Python?
A. function
B. def
C. define

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C. define
D. fun
Enter your answer (A/B/C/D): B

Q 2 . Which data type stores multiple values in Python?
A. list
B. int
C. float
D. bool
Enter your answer (A/B/C/D): A

```

```
Q 3 . Which symbol is used for comments in Python?  
A. //  
B. /*  
C. #  
D. --  
Enter your answer (A/B/C/D): C
```

```
Q 4 . Which function displays output in Python?  
A. input()  
B. print()  
C. display()  
D. output()  
Enter your answer (A/B/C/D): B
```

```
Q 5 . Which keyword is used to loop through a sequence?  
A. repeat  
B. loop  
C. for  
D. next  
Enter your answer (A/B/C/D): C
```

FINAL RESULT & OUTPUT:

```
Missing Answers: 0  
Score: 5 / 5  
Percentage: 100.00%  
Grade: A+  
=====  
Quiz completed successfully!
```

```
===== RESULT REPORT =====  
Student Name: SIVARANGJENII  
Date: 25-08-2026 04:09:42  
Total Questions: 5  
Correct Answers: 5  
Wrong Answers: 0  
Missing Answers: 0
```